1. Executive Summary & Core Value

CCS Forge is designed to transform the student experience in the College of Computer Studies (CCS) from a fragmented series of chat messages into a unified, gamified ecosystem. By centralizing events, challenges, and project showcases, it creates a "digital home" for CS and IT students to build a visible record of their growth.

---

2. Technical Architecture: The Monorepo Strategy

Using **Turborepo** allows you to manage multiple applications and shared logic in one place, ensuring high developer velocity for your student team.

- **apps/web (The Content Hub)**: Built with **Astro**. This handles the Landing Page and the **Resource Repository**. Since these are content-heavy, Astro delivers them as pure HTML for instant loading on mobile devices.
- **apps/dashboard (The Interactive Portal)**: An **Astro SSR** app using **React Islands**. This is where students track points, join challenges, and view their personalized progress snapshots.
- **packages/database (The Single Source of Truth)**: Contains the **Prisma** schema. This ensures that both the dashboard and the landing page use the exact same TypeScript types for "Challenges" and "Users," preventing data bugs.

---

3. Detailed Feature Deep-Dive

The Challenge System (Core Differentiator)

The "Forge" isn't just a list of tasks; it’s a gamified engine designed to increase active learning.

- **Flexible Modalities**: Supports coding challenges, quiz battles, and team-based scavenger hunts.
- **Prisma Integration**: Every submission is tracked in the database with fields for difficulty, points, and evaluation criteria.
- **Verification**: Faculty and officers can review submissions and approve points directly through the admin panel.

Gamification & Recognition

This module converts student effort into digital capital.

- **Dynamic Leaderboards**: Using Prisma to aggregate points across individual and team challenges to display rankings.
- **Badge Engine**: Automated awarding of digital badges for milestones like "First Code Submission" or "Event Streak".

Project Showcase

A public-facing gallery where students can publish capstone outputs and UI/UX research.

- **Visibility**: Designed to make student skills visible to peers and potential recruiters.
- **Performance**: Astro generates these pages statically, making student portfolios SEO-friendly and fast.

---

4. Operational Roles & Security

The system uses **Role-Based Access Control (RBAC)** to ensure data integrity.

Role	Responsibility	Data Access (Prisma/Supabase)
Student	Participation and Submissions	Can only modify their own Submission and Profile records.
Student Officer	Activity Management	Can create/edit Event and Challenge records for their organization.
Faculty/Admin	Moderation and Analytics	Access to all participation reports and system configuration.

---

5. Success Metrics & Roadmap

To avoid "scope creep," the MVP focuses on the minimum features needed to show value.

Success Metrics

- **Adoption**: Number of monthly active users within the department.
- **Completion**: Ratio of challenge participation to successful completions.
- **Retention**: The "Weekly Return Rate" of students checking for new announcements.

The Release Path

1. **Phase 1: Foundation**: Set up the monorepo, define the Prisma schema, and configure Supabase Auth.
2. **Phase 2: Core Loop**: Deploy the Dashboard and the first set of "Attendance Quests" and "Announcements".
3. **Phase 3: Forge Launch**: Enable the full Challenge module and Gamification points.
4. **Phase 4: Advanced Features**: Roll out team-based challenges and the public Project Showcase.